



We may study various factors and collect information on different subjects in our daily life.

Let us consider some of these subjects.

1. Number of girls in different classes in a school
2. Number of boys of a class, playing different games
3. Number of cows in different villages
4. Number of toys with different boys
5. Number of different animals in a zoo



We may collect information on all these subjects by counting.

The information on subject 1 will have names of different classes, and the number of girls in each of them.

The information on subject 2 will have names of different games, and the number of boys playing each of them.

The information on subject 3 will have names of different villages, and the number of cows in each of them.

The information on subject 4 will have names of different boys, and the number of toys that each of them has.

The information on subject 5 will have names of different animals, and the number of each of them in the zoo.

Such information, which we collect on a subject, are called data.

Thus, we define **data** as a collection of facts and figures.

We may represent a data by tables, pictures or graphs.

When we make use of picture symbols to represent information, we call it pictorial representation or **pictograph** of the given information (data).

Drawing a Pictograph for a Given Information

Given below is the information about the number of boys, who like different fruits.

Apple	Orange	Banana	Pear	Guava
6	4	2	7	1

Let us draw a pictograph for this data. The numbers in the given table show the number of boys. So, we use the figure of a boy to represent the information.

We get a pictograph as shown below.

Number of Boys Liking Different Fruits

Apple	     
Orange	   
Banana	 
Pear	      
Guava	

Now, consider the following information about the number of absentees (boys and girls) in a class on various days of a week.

Days	Number of Absentees	
	Boys	Girls
Monday	1	2
Tuesday	2	0
Wednesday	1	1
Thursday	2	1
Friday	0	2

Let us use the figure of the face of a boy to represent a boy and the face of a girl to represent a girl.

We, then, get a pictograph as shown on next page.

Number of Absentees on Various Days of a Week

Monday	  
Tuesday	 
Wednesday	 
Thursday	  
Friday	 

Sometimes, the numbers in the data are large and it is not possible to draw so many figures.

In such cases, we use one picture or symbol to represent a specific number of objects. The number of persons or objects that each picture or symbol represents is indicated as the key just below the pictograph.

Given below is the information about 240 boys of a school, who play different games.

Football	Cricket	Volley-ball	Basket-ball	Badminton
60	80	30	50	20

It is difficult to draw 60 or 80 figures. So, we use one figure to represent 10 boys.

Thus, we have the pictograph shown below.

Number of Boys Playing Different Games

Football	     
Cricket	       
Volley-ball	  
Basket-ball	    
Badminton	 

One  represents 10 boys.



Exercise 67

1. Given below is the information about number of toys owned by five students.

Sajal	Anshuman	Kartik	Preeti	Sanjana
8	2	7	4	6

Represent the above information by a pictograph.

2. The following information shows the number of different kinds of animals in a zoo.

Monkey	Giraffe	Lion	Elephant	Deer
9	2	3	6	5

Represent the above data by a pictograph.

3. The following information shows the number of different kinds of vehicles in a parking-lot.

Buses	Cars	Scooters	Motorcycles	Bicycles
10	60	20	40	30

Represent the above data by a pictograph.

4. The table given below shows the number of bottles of different brands of cold-drinks sold by a shopkeeper on a certain day.

Fanta	Slice	Coke	Limca	Sprite
15	25	30	5	20

Represent the above data by a pictograph.

Reading and Interpreting a Pictograph

Study the following examples.

Example 1: The pictograph given below shows the number of letters received by Mr Roy on various days of a week. Read the pictograph and answer the following questions.

Monday	
Tuesday	
Wednesday	

Thursday	
Friday	 
Saturday	    

- (a) On which day did Mr Roy receive the maximum number of letters?
 (b) On which day did Mr Roy receive no letter?
 (c) How many letters were received by Mr Roy on Saturday?
 (d) How many letters were received by Mr Roy during the whole week?

Solution:

- (a) Seeing the pictograph, we find that the maximum number of letters received by Mr Roy is 8, on Wednesday.

So, Mr Roy received the maximum number of letters on Wednesday.

- (b) In the pictograph, we can observe that no letter has been shown against Thursday.

So, Mr Roy did not receive any letter on Thursday.

- (c) In the pictograph, 5 letters have been shown against Saturday.

So, Mr Roy received 5 letters on Saturday.

- (d) Clearly, total number of letters received by Mr Roy during the whole week

$$\begin{array}{cccccc} \text{Mon.} & \text{Tue.} & \text{Wed.} & \text{Thu.} & \text{Fri.} & \text{Sat.} \\ = & 6 & + & 3 & + & 8 & + & 0 & + & 2 & + & 5 & = & 24. \end{array}$$

Example 2: Below is given a pictograph showing the number of students of a class liking different food items.

Pizza	    
Dosa	
Burger	     
Pasta	 
Sandwich	   
Chips	  

One  represents 2 students.

Study the given pictograph on previous page and answer the following questions.

- (a) How many students like Pizza?
- (b) Which food item is most popular among the students of the class? How many students like it?
- (c) Which food item is least popular ? How many students like it?
- (d) Which food item is liked by 8 students only?
- (e) How many students are there in the class?

Solution:

- (a) Clearly, number of students who like pizza

$$= (\text{Number of } \text{😊}) \times 2 = 5 \times 2 = 10.$$

- (b) The most popular food item will be the one which has maximum number of 😊.

Such a food item is Burger which has six 😊.

$$\text{Number of students who like Burger} = (6 \times 2) = 12.$$

∴ The most popular food item is Burger and it is liked by 12 students.

- (c) The least popular food item will be the one which has least number of 😊.

Such a food item is Dosa, which has one 😊.

$$\text{Number of students who like Dosa} = (1 \times 2) = 2.$$

∴ The least popular food item is Dosa and it is liked by 2 students.

- (d) 8 students are represented by $(8 \div 2) = 4$ 😊.

The food item having four 😊 is Sandwich.

∴ Sandwich is liked by 8 students only.

- (e) Number of students in the class

$$= (\text{Total number of } \text{😊} \text{ in the pictograph}) \times 2$$

$$= (5 + 1 + 6 + 2 + 4 + 3) \times 2$$

$$= 21 \times 2 = 42.$$



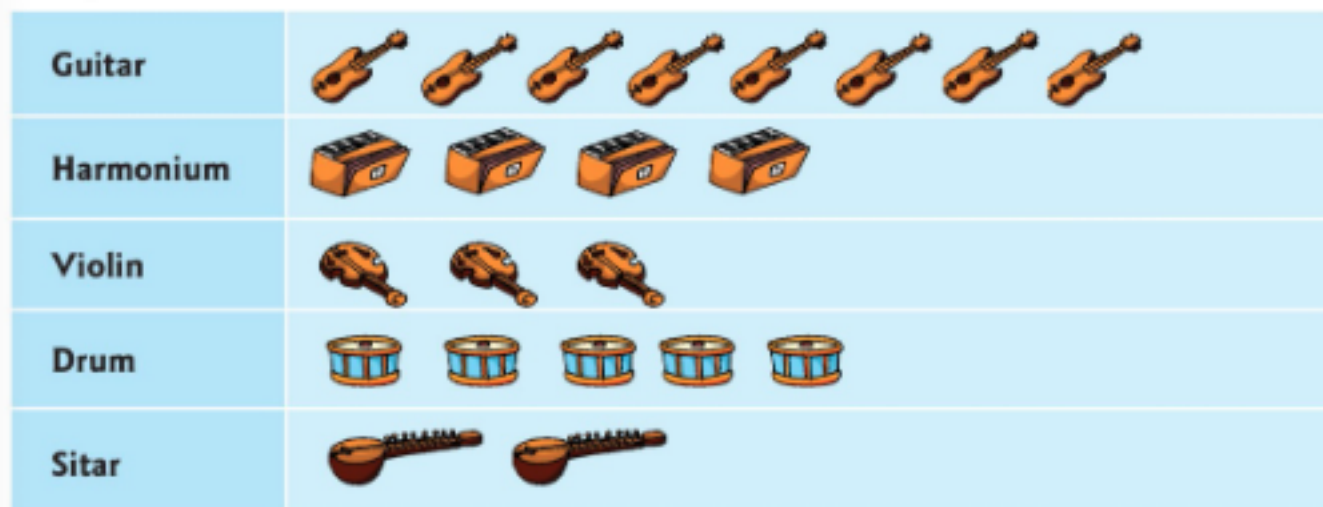
Exercise 68

1. The following pictograph shows the number of girls who are fond of reading different fairy-tales.



Read the above pictograph and answer the questions given below:

- Which fairy-tale is most popular among the girls?
 - How many girls love to read Cinderella?
 - Which fairy-tale is least liked by the girls?
2. The following pictograph shows the number of different kinds of musical instruments in a shop.



Read the above pictograph and answer the questions given below:

- Which musical instrument is in greatest demand?
- The number of which two musical instruments together equal the number of guitars in the shop?
- Which musical instrument is available in least number in the shop?

3. The pictograph shown below shows the number of students in a library, reading different kinds of books.



Read the above pictograph and answer the questions:

- How many students are there in the library?
 - What is the difference between the number of students reading comics and those reading novels?
 - Which type of books is read by the maximum number of students?
4. The following pictograph shows the number of fruits bought by a fruit vendor from a wholesale merchant.



Each picture represents 5 fruits

Study the above pictograph carefully and answer the following questions.

- (a) How many different kinds of fruits did the vendor buy?
- (b) Which fruit did he buy in least number?
- (c) How many bananas did the vendor buy?
- (d) How many guavas did he buy?
- (e) How many more apples did he buy than mangoes?
- (f) How many fruits did he buy in all?

Collecting Data Using Tally Marks

Suppose we have a row of 30 boys, who like different games— Cricket, Hockey, Football, Chess and Badminton.

We go on asking each boy his choice of games and recording the collected information.

In such a case, we prepare a table containing the available choices.

Then, for each boy we mark a vertical line called tally mark against his particular choice.

Once we have a set of four tally marks for a certain choice, then we indicate the fifth mark by crossing the four tally marks. We then start with a new set of tally marks.

Suppose the number of boys liking different games are as given below:

Cricket	Hockey	Football	Chess	Badminton
8	5	7	6	4

The above information may be collected as under:

Game	Tally marks	Number of boys
Cricket		8
Hockey		5
Football		7
Chess		6
Badminton		4



Activity Time



1. Take a die and throw it 40 times. Record your output at each throw of the die, by drawing a tally mark in front of the number of dots you get. Make a table as shown below:

Face of the die	Tally Marks	Number of times
		
		
		
		
		
		

- (a) Which face of the die did you get the most number of times?
(b) Which face of the die did you get the least number of times?
2. Ask each of your classmates about the mode of transport he/she uses to come to school. Then make a table as in the above activity, using the heads: Mode of transport, Tally marks and number of students.
- (a) Which is the most popular mode of transport among your classmates?
(b) Which mode of transport is the least popular?

